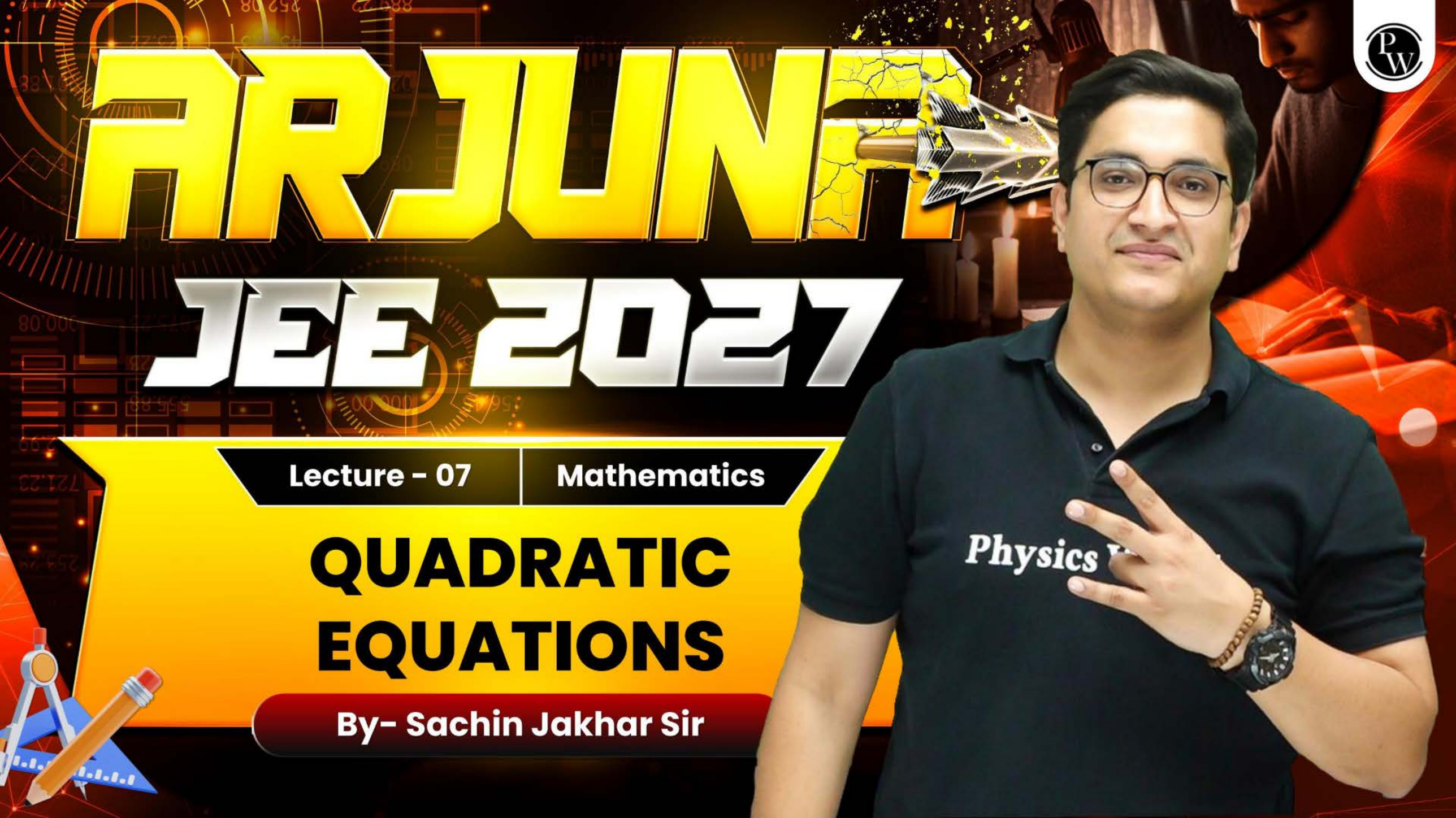
ARJUNA JEE 2027

Lecture - 07 | Mathematics

QUADRATIC EQUATIONS

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Topics *to be covered*

A Graph of Quadratic

B Range of Quadratic

C Question Practice

D





LAST CLASS REVISION

Answer the following Questions:

(1) If one root of quadratic is $p+\sqrt{q}$ then other root must be $p-\sqrt{q}$. → (f)

(2) Quadratic can have a irrational & a rational root. → (T)

(3) Quadratic with rational coefficient can have a irrational & a rational root. → (f)

(4) If Quadratic with real coefficients have a imaginary root $p+iq$ then other root is its conjugate $p-iq$. → (T)

LAST CLASS REVISION

Sum of coeff $= 3\alpha^2 - \beta^2 - \alpha\beta - 2\alpha^2 + \beta^2 - \alpha^2 + \alpha\beta$
$= 0.$

Answer the following Questions:

5. For quadratic : $(3\alpha^2 - \beta^2 - \alpha\beta)x^2 - 2\alpha^2 x + (\beta^2 - \alpha^2 + \alpha\beta) = 0$ roots are _____.

$1, \left(\frac{\beta^2 - \alpha^2 + \alpha\beta}{3\alpha^2 - \beta^2 - \alpha\beta}\right)$

6. Quadratic with rational coefficient & whose one roots is $\sqrt{7} + 3$, is _____.

$\{3 + \sqrt{7}, 3 - \sqrt{7}\}$

$x^2 - (6)x + 2 = 0$

** $x = 3 + \sqrt{7}$

SBS $x - 3 = \sqrt{7}$

$x^2 + 9 - 6x = 7$

$x^2 - 6x + 2 = 0$

DIBY-35

If $x = 3 + \sqrt{5}$ then the value of $x^4 - 12x^3 + 44x^2 - 48x + 17$.

(A) −1 (B) 0 (C) 1 (D) 2

[Ans. (C)]

Quad. with Rat coeff

$x - 3 = \sqrt{5}$

SBS.

$x^2 - 6x + 9 = 5$

$x^2 - 6x + 4 = 0$

$3 + \sqrt{5}$

$3 - \sqrt{5}$

$x^2 - 6x + 4 \overline{)x^4 - 12x^3 + 44x^2 - 48x + 17} \; (x^2 - 6x + 4)$

$x^4 - 6x^3 + 4x^2$ (− + −)

$-6x^3 + 40x^2 - 48x$

$-6x^3 + 36x^2 - 24x$ (+ − +)

$4x^2 - 24x + 17$

$4x^2 - 24x + 16$ (− + −)

1

$B \overline{)A} \; Q$, R

$A = BQ + R$

$$x^4-12x^3+44x^2-48x+17 = (x^2-6x+4)(x^2-6x+4)+1$$

$$x = 3+\sqrt{5}$$

Question

If $x = \sqrt{11} + 4$ then value of $x^5 - 8x^4 + 12x^3 - 8x^2 + 11x + 5$ is _____.

[Ans. : 5]

QIBY - (A)

Question Practice

Acidic Question

If $p(q-r)x^2 + q(r-p)x + r(p-q) = 0$ has equal roots prove $\frac{2}{q} = \frac{1}{p} + \frac{1}{r}$

1

$\frac{c}{a}$ $\quad \frac{c}{a} = 1$

M-2

$D = 0$

$m \quad n$

$$(q(r-p))^2 - 4\,p(q-r)\,r(p-q) = 0$$

$$(-(m+n))^2 - 4mn = 0$$

$-2mn$

$$m^2 + n^2 + 2mn - 4mn = 0$$

$$(m-n)^2 = 0$$

$\# \; m = n$

$m = n$

$$pq - pr = rp - rq$$

$$\frac{pq}{pqr} + \frac{rq}{pqr} = \frac{2rp}{pqr}$$

$$\frac{1}{r} + \frac{1}{p} = \frac{2}{q}$$

$$m = p(q-r) = pq - pr$$

$$n = r(p-q) = rp - rq$$

add

$$m+n = pq - rq$$

$$= q(p-r)$$

$$-(m+n) = q(r-p)$$

Question

If $a, b, c \in Q$, then roots of the equation $(b+c-2a)x^2 + (c+a-2b)x + (a+b-2c) = 0$ are

(1) $\frac{a+b-2c}{b+c-2a}$

$m = b+c-2a$

$n = a+b-2c$

$m+n = 2b-a-c$

$= -(a+c-2b)$

If D = perfect square

A. Rational

B. Non-real

C. Irrational

D. Equal

$$D = (c+a-2b)^2 - 4\underbrace{(b+c-2a)}_{m}\underbrace{(a+b-2c)}_{n} = (\quad)^2$$

$$D = (-(m+n))^2 - 4mn$$

$$= m^2+n^2+2mn-4mn$$

$$= (m-n)^2 = (b+c-2a-a-b+2c)^2 = (3c-3a)^2$$

Challenger Question

Let a, b, c, d are positive real numbers such that $\frac{a}{b} \neq \frac{c}{d}$, then the roots of the equation $(a^2+b^2)x^2 + 2x(ac+bd) + (c^2+d^2) = 0$ are:

A real and distinct

B real and equal

C imaginary

D nothing can be said

D = -ve → Imag Roots

$$D = (2(ac+bd))^2 - 4(a^2+b^2)(c^2+d^2)$$

$$= 4(a^2c^2+b^2d^2+2acbd) - 4(a^2c^2+a^2d^2+b^2c^2+b^2d^2)$$

$$= 4\{a^2c^2+b^2d^2+2acbd - a^2c^2 - a^2d^2 - b^2c^2 - b^2d^2\}$$

$$= -4\{(ad)^2+(bc)^2-2(ad)(bc)\} = \underbrace{-4}_{\text{-ve}}\underbrace{(ad-bc)^2}_{\text{+ve}} = \text{-ve always}$$

Critical Thinking Question (CTQ)

If both the roots of the quadratic equation $x^2 - (2n + 18)x - n - 11 = 0, n \in I$, are rational, then find the value(s) of n.

Re-think.

$n = -8, -11$

Question [JEE Mains-2022]

If for some $p, q, r \in R$, not all have same sign, one of the roots of the equation $(p^2+q^2)x^2 - 2q(p+r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{q^2+r^2}{p^2}$ is equal to

$$\frac{q^2+r^2}{p^2} = \frac{(-4p)^2+(16p)^2}{p^2} = \frac{272p^2}{p^2} = 272 \checkmark$$

p	q	r	
+	+	−	
+	−	+	
−	+	+	
−	−	+	
−	+	−	
+	−	−	
+	+	+	X
−	−	−	X

$$x^2p^2 - 2qpx + q^2 + x^2q^2 - 2rqx + r^2 = 0$$

$$(xp-q)^2 + (xq-r)^2 = 0$$

($(xp-q)$: ≥ 0, $(xq-r)$: ≥ 0)

$xp - q = 0$ & $xq - r = 0$.

$x = \frac{q}{p}$ & $x = \frac{r}{q} \Rightarrow \boxed{\frac{q}{p} = \frac{r}{q}}$

$x^2 + 2x - 8 = 0$

$(x+4)(x-2) = 0$ → 2, −4

$\frac{q}{p} = \frac{r}{q} = 2 \Rightarrow q = 2p,\ r = 2q$ → p, q, r same sign. ✗

$\frac{q}{p} = \frac{r}{q} = -4$ ✓

$\boxed{q = -4p \quad r = -4q}$

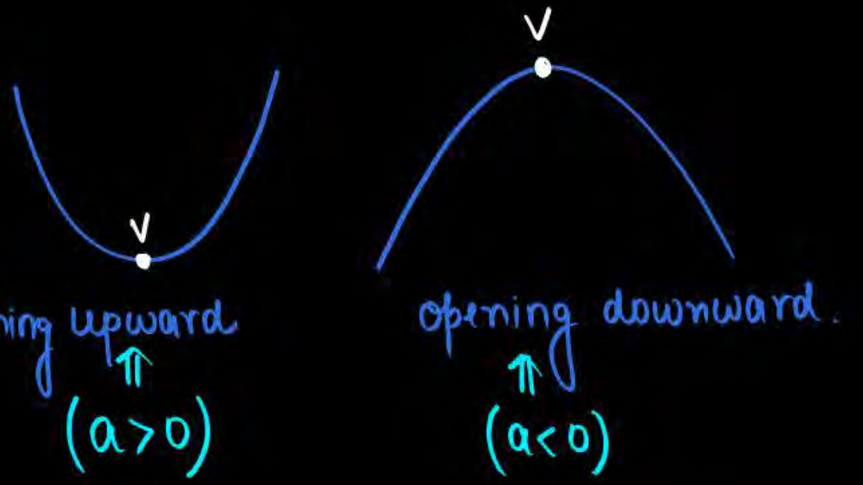
p	q	r
−	+	−
+	−	+

$r = -4(-4p) = 16p.$



GRAPH OF QUADRATIC

Parabolic



$$f(x) = y = ax^2 + bx + c$$

** Vertex $\equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$

x-cord. y-cord.

put $x = 0 \Rightarrow y = c$

$(0, c)$ is the point where graph cuts y-axis

$$f(x) = y = ax^2 + bx + c$$

$$= a\left(x^2 + 2\cdot\frac{b}{2a}x + \frac{b^2}{4a^2} + \frac{c}{a} - \frac{b^2}{4a^2}\right)$$

$$\# \quad y = a\left(x + \frac{b}{2a}\right)^2 - \left(\frac{b^2 - 4ac}{4a}\right)$$

$$x = -\frac{b}{2a} \Rightarrow y = -\frac{D}{4a}.$$

$a > 0$

min. value

$$y = -\frac{D}{4a}.$$

$a < 0$.

$$y_{max.} = -\frac{D}{4a}$$



• Imag. Roots

ex:-

$y = x^2 + x + 1$

$a = 1, b = 1, c = 1, D = 1^2 - 4 \times 1 \times 1$

$= -3$

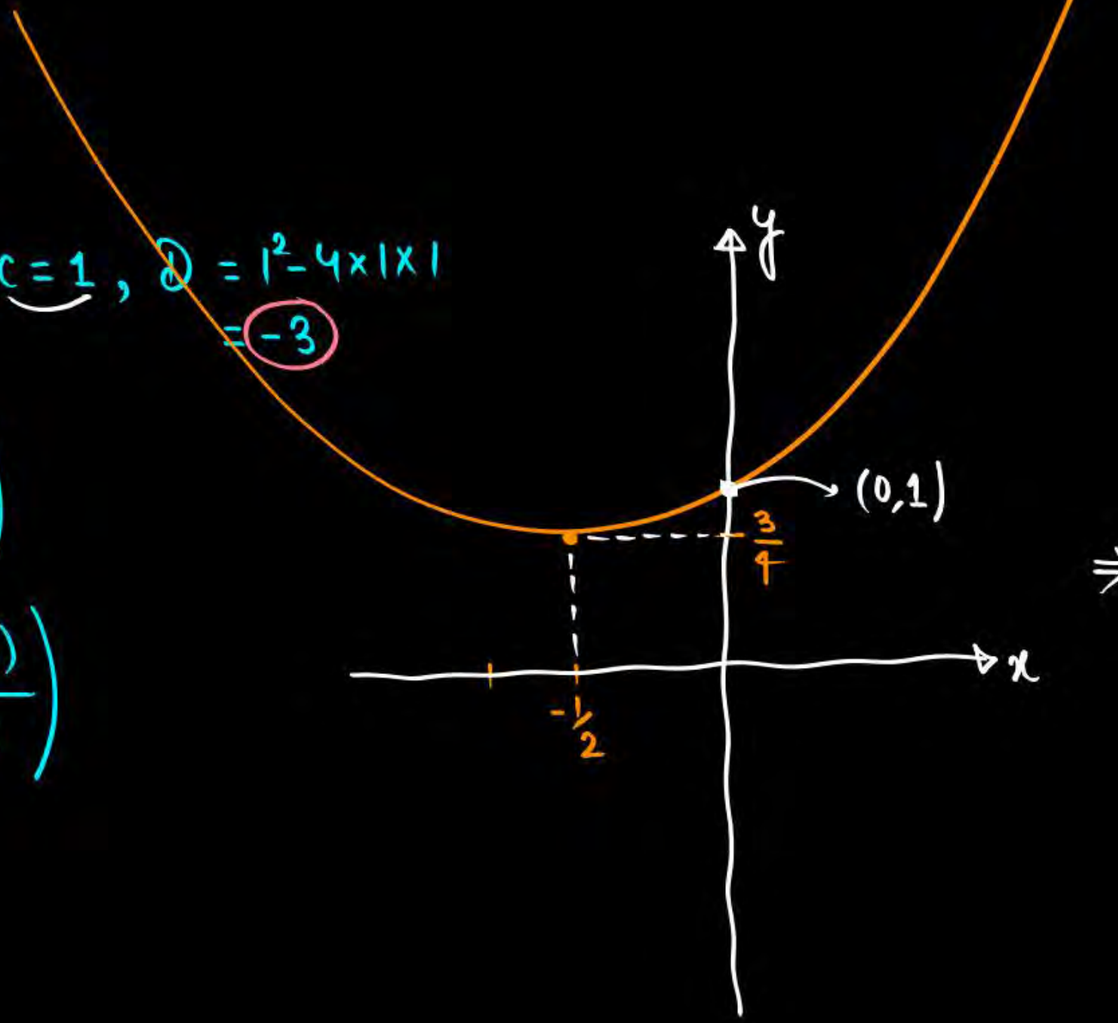
$a \equiv +ve$



$$V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$$

$$\equiv \left(\frac{-1}{2\times 1}, \frac{-(-3)}{4\times 1}\right)$$

$$V \equiv \left(\frac{-1}{2}, \frac{3}{4}\right)$$



$\Rightarrow$ Graph never intersects x-axis

$\Downarrow$

No Real Roots.

$y = x^2 - 7x + 12$

$a = 1,\ b = -7,\ c = 12,\ D = 49 - 4 \times 1 \times 12$

$D = 1$

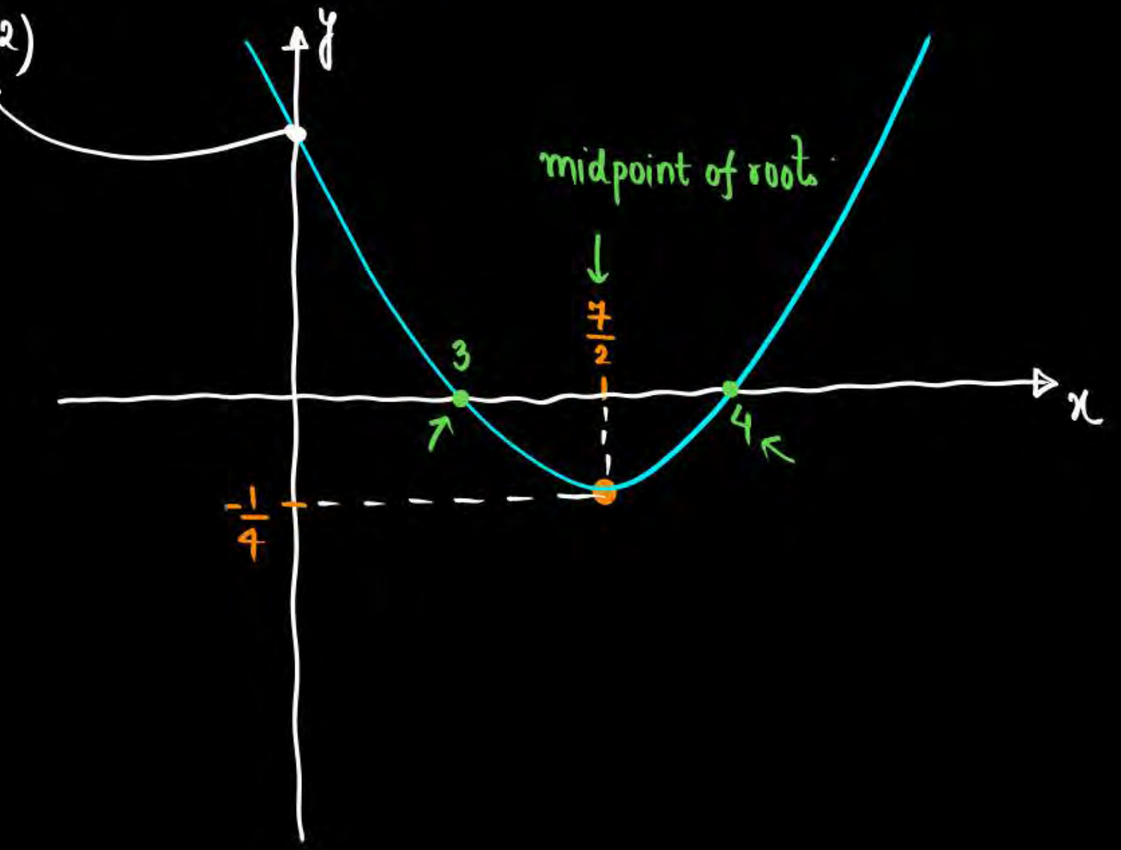
$V \equiv \left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

$\left(\frac{-(-7)}{2\times1}, \frac{-(1)}{4\times1}\right)$

$V = \left(\frac{7}{2}, -\frac{1}{4}\right)$

$x^2 - 7x + 12 = 0$ → 3, 4

$(0, 12)$



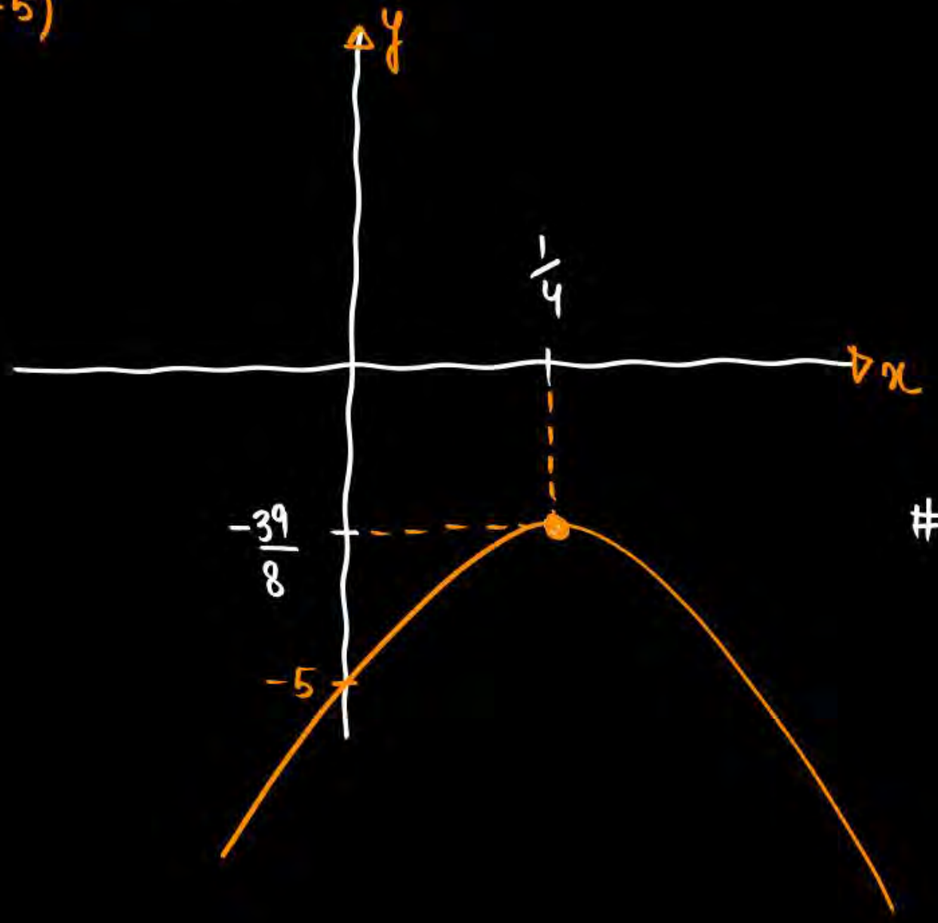
$y = -2x^2 + x - 5$, $a = -2$, $b = 1$, $c = -5$, $D = -39$

$(0, -5)$

$$V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$$

$$\equiv \left(\frac{+(1)}{2\times+2}, \frac{+(+39)}{4(-2)}\right)$$

$$V = \left(\frac{1}{4}, \frac{-39}{8}\right)$$



No Real Root.

$y = x^2 + 6x + 9$

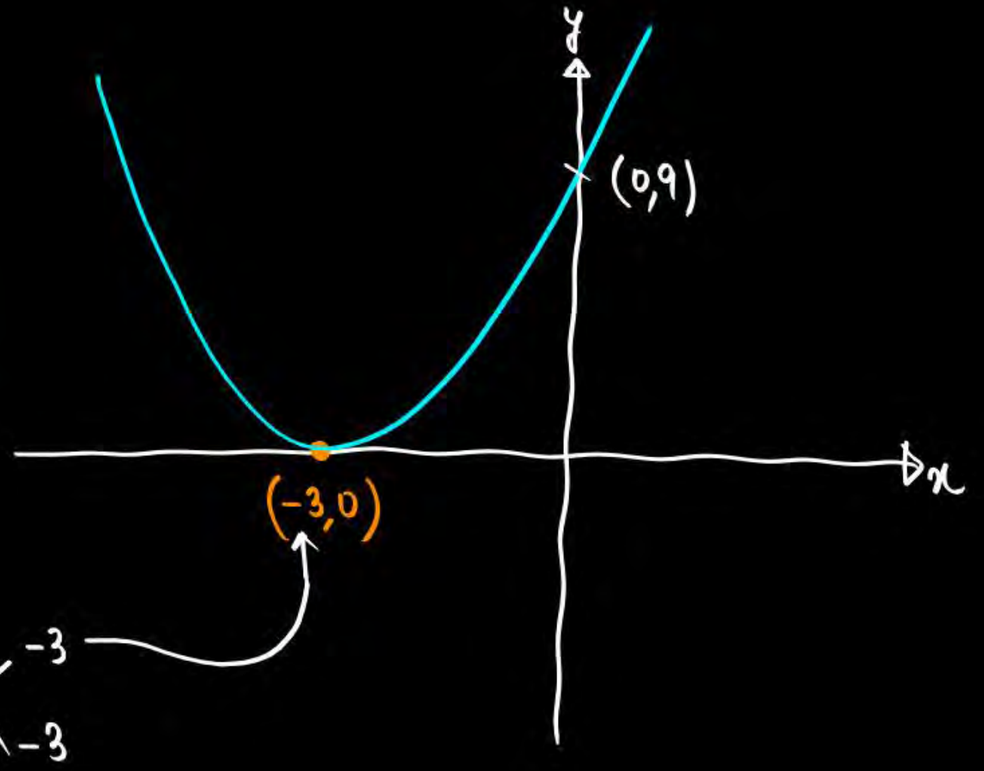


$a = 1,\ b = 6,\ c = 9,\ D = 0.$

$$V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$$

$$\equiv \left(\frac{-6}{2\times 1}, \frac{-0}{4\times 1}\right)$$

$$\equiv (-3, 0)$$



$x^2 + 6x + 9 = 0$ → $-3, -3$

equal Roots.

Question

Draw graph of following quadratic :

(i) $f(x) = x^2 + x - 12$

(ii) $f(x) = x - (1 + x^2)$

(iii) $f(x) = x^2 - 10x + 16$

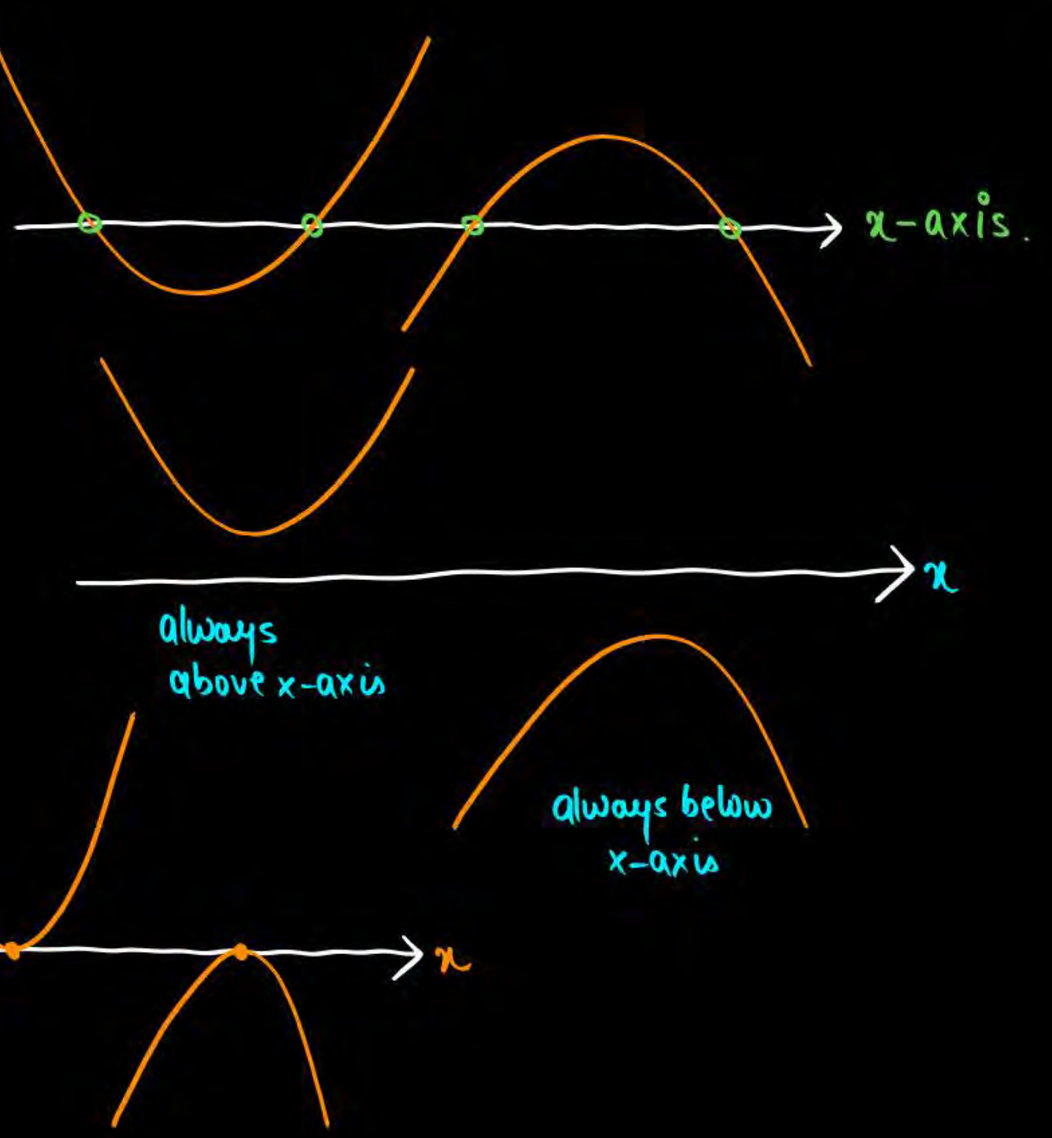
QIBY - (B)

DISCRIMINANT & GRAPH

$D > 0 \Rightarrow$ Real & Distinct Roots.

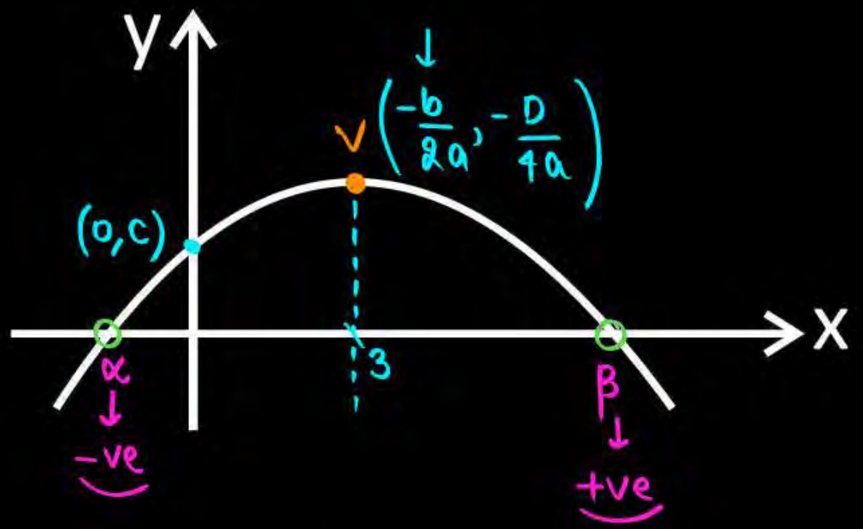
$D < 0 \Rightarrow$ Imag. (No Real) Roots.

$D = 0 \Rightarrow$ Roots are Real & equal.



Question

Comment on sign of a, b, c & D for following graphs:



opening downward → $a < 0$

graph cuts x-axis at 2 dist. points → $D > 0$

graph cut +ve y-axis ⇒ $c > 0$

$\frac{-b}{a}$ = Sum of Roots ⇒ +ve

+ve ← $\frac{c}{a}$ = Product of Roots ⇒ -ve

-ve ← a

$b > 0$

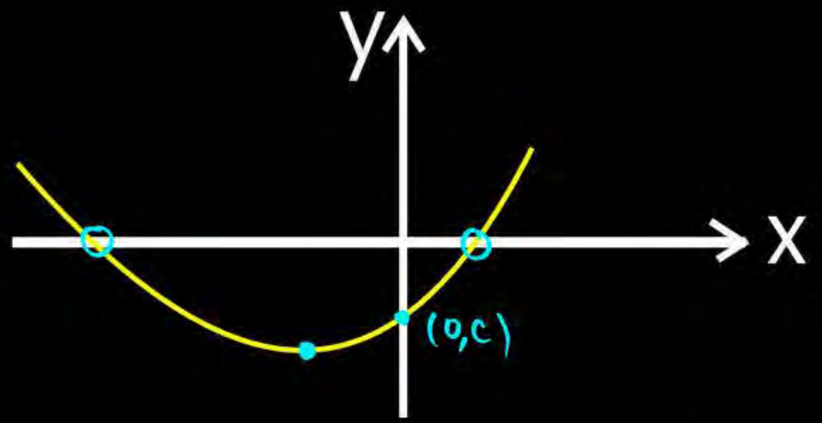
$\Rightarrow \frac{-b}{2a} > 0$

$\frac{-b}{a} > 0 \Rightarrow -b < 0$

$b > 0$

Question

Comment on sign of a, b, c & D for following graphs:



$a > 0$

$D > 0$

$c < 0$

$b > 0$

$S = \frac{-b}{a} \Rightarrow$ (-ve)

$P = \frac{c}{a} \Rightarrow$ (-ve)

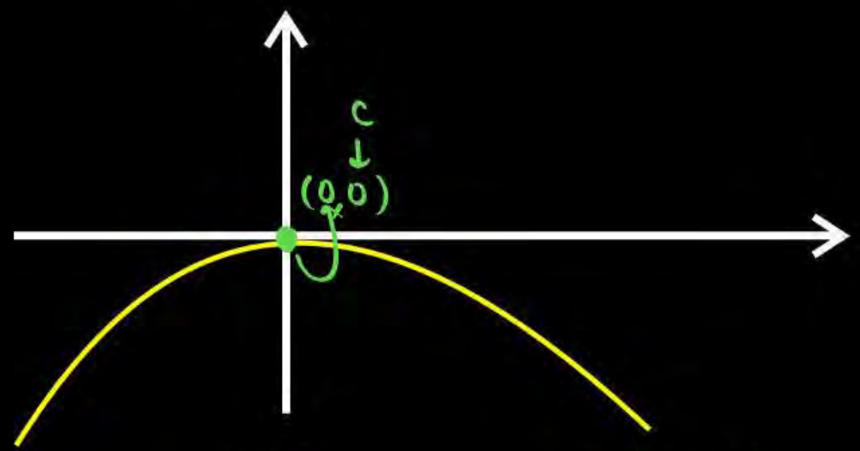
vertex ka x-cord. = III quad.

$\frac{-b}{2a} < 0$

2a → +ve

$-b < 0$

$b > 0$.



$a < 0$.

$c = 0$

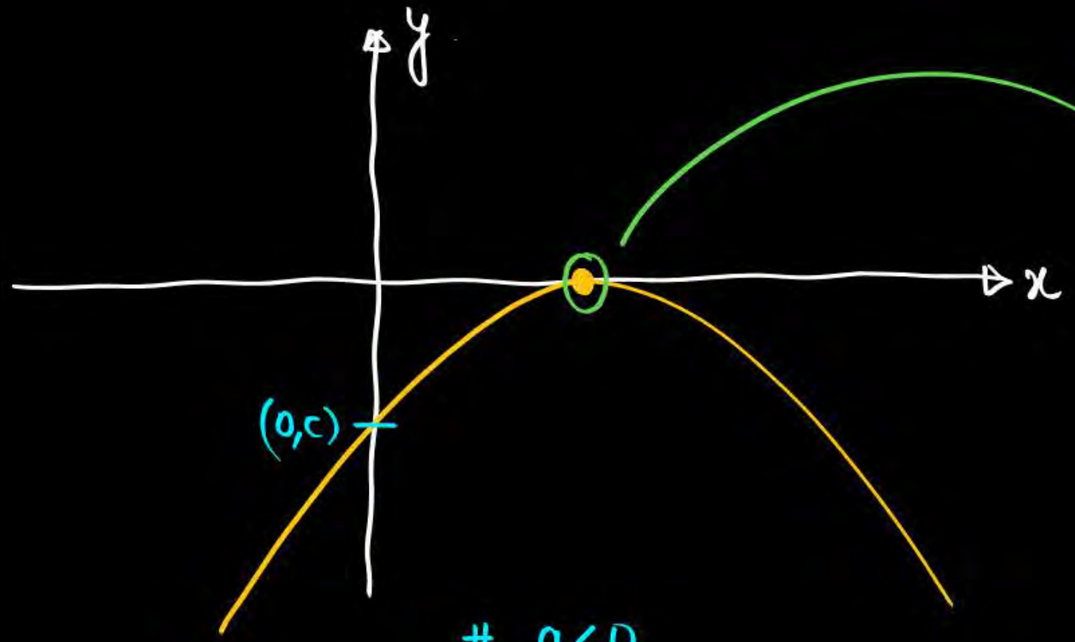
$D = 0$.

$b = 0$.

$S = 0$

$P = 0$.

$-\frac{b}{2a} = 0$



$-\frac{b}{2a} > 0$

$2a$ → -ve

$-b < 0$

$b > 0$

$a < 0$

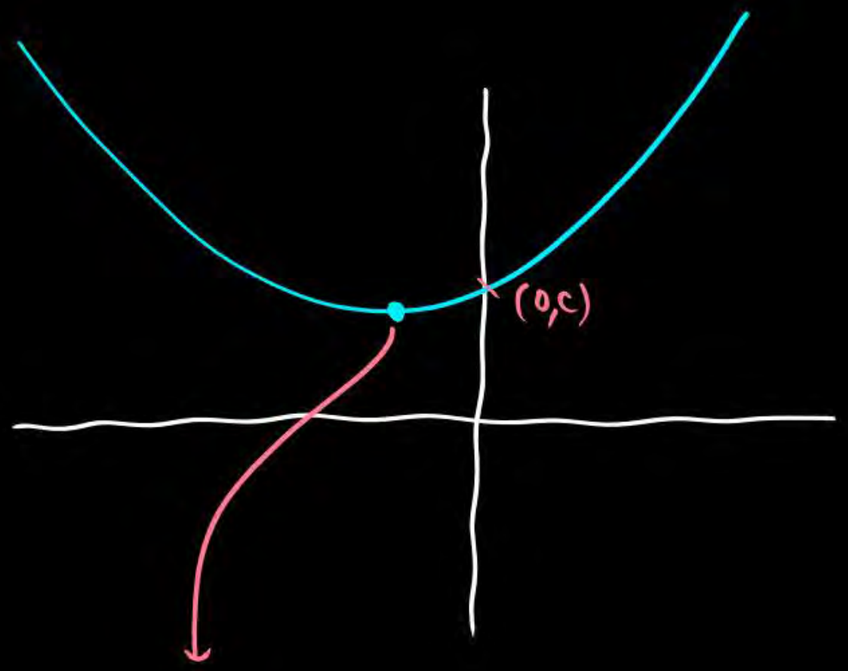
$c < 0$

$D = 0$

$b > 0$

$S = -\frac{b}{a}$ → (+ve)

$P = \frac{c}{a}$ → (+ve)



$a > 0$

$c > 0$

$D < 0$

$b > 0$

$\xi = -\frac{b}{a} \rightarrow \ominus$

$P = \frac{c}{a} \rightarrow \oplus$

$\frac{-b}{2a} < 0$

$-b < 0$

$b > 0$

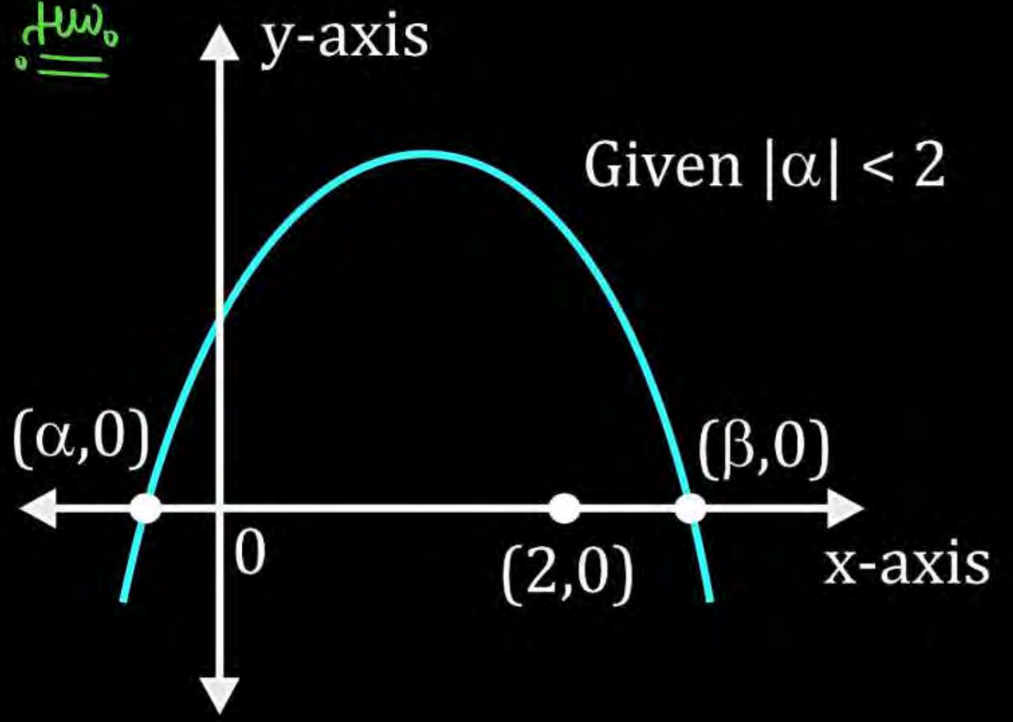
Acidic Question

The graph of $y = ax^2 + bx + c$ is shown in the figure, then which of the following is(are) correct?

A $ab^2c^3 < 0$

B $ab < 0$

C $bc(4a + 2b + c) > 0$



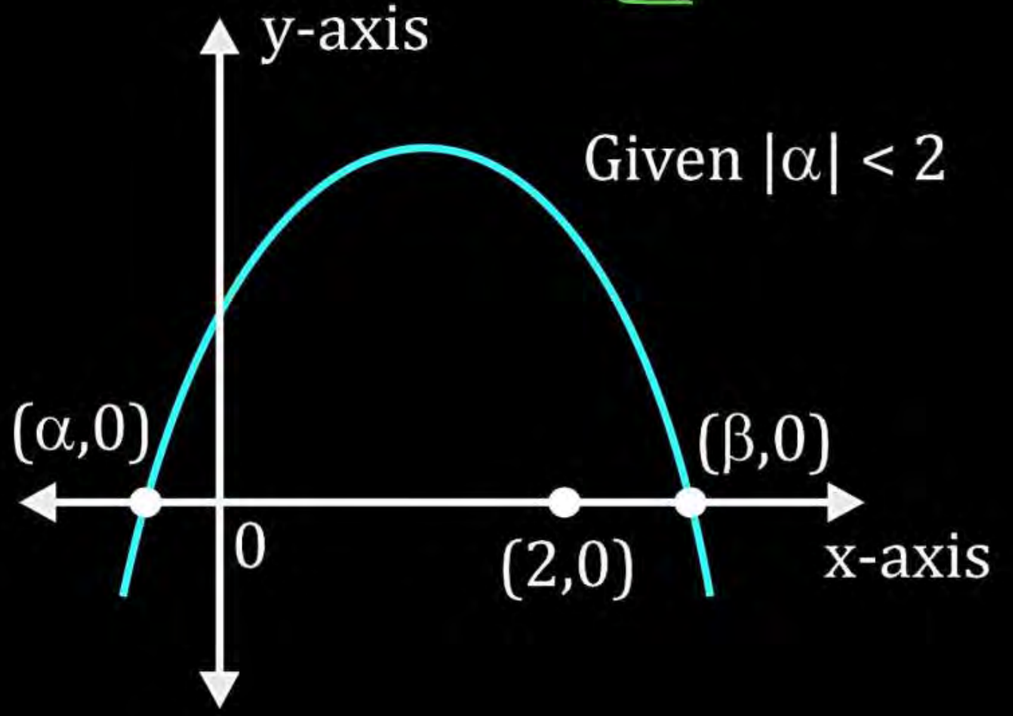
Acidic Question

The graph of $y = ax^2 + bx + c$ is shown in the figure, then which of the following is(are) correct?

D $ab(4a - 2b + c) > 0$

E $b^2 > 4(a^2 + ac)$

F $ax^2 + bx + c > 0 \; \forall \; x \in (0, \beta)$



DO IT BY YOURSELF (DIBY)

DIBY-33

If the equation $\frac{x^2-bx}{ax-c}=\frac{m-1}{m+1}$ has roots equal in magnitude but opposite in sign, then m is

(A) $\frac{a+b}{a-b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{b-a}{b+a}$ (D) None of these

[Ans. : (B)]

DIBY-34

Find the value of a for which one root of the equation $x^2+(2a-1)x+a^2+2=0$ is twice as large as the other.

[Ans. : $a=-4$]

DIBY-35

If $x=3+\sqrt{5}$ then the value of $x^4-12x^3+44x^2-48x+17$. ✓

(A) –1 (B) 0 (C) 1 (D) 2

[Ans. (C)]

DIBY-36

The quadratic equation $x^2 + ax + b + 1 = 0$ has both positive integral roots, then $a^2 + b^2$ can be equal to

(A) 50 (B) 37 (C) 61 (D) 20

[Ans. : A, D]

DIBY-37

The number of real roots of the equation $5 + |2^x - 1| = 2^x(2^x - 2)$ is

[JEE Mains-2019]

(A) 4 (B) 3 (C) 2 (D) 1

[Ans. : (D)]

DIBY-38

Find the product of the real roots of the equation, $\overbrace{x^2 + 18x + 30}^{t} = 2\sqrt{x^2 + 18x + 45}$

[Ans. : 20]

DIBY-39

If 'α' is an irrational solution of the equation $3^{2x^2-7x+3} = 4^{x^2-x-6}$, then the value of $3^{\left(\frac{2\alpha-1}{\alpha+2}\right)}$ is

[Ans. : 4]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians